



US012416981B2

(12) **United States Patent**
Lee

(10) **Patent No.:** **US 12,416,981 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **COMPUTER MOUSE**

(71) Applicant: **Han Kiel Lee**, Seoul (KR)

(72) Inventor: **Han Kiel Lee**, Seoul (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/929,628**

(22) Filed: **Oct. 29, 2024**

(65) **Prior Publication Data**

US 2025/0053248 A1 Feb. 13, 2025

Related U.S. Application Data

(62) Division of application No. 18/261,676, filed on Jul. 17, 2023, now Pat. No. 12,158,997.

(51) **Int. Cl.**
G06F 3/0354 (2013.01)
G06F 3/0362 (2013.01)

(52) **U.S. Cl.**
CPC **G06F 3/03543** (2013.01); **G06F 3/0362**
(2013.01)

(58) **Field of Classification Search**
CPC G06F 3/03543; G06F 3/0362; G06F
2203/0333
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,260,696 A * 11/1993 Maynard, Jr. G06F 3/039
345/157
D372,906 S * 8/1996 Kawauchi D14/402

5,771,038 A * 6/1998 Wang G06F 3/03543
345/161
5,847,696 A * 12/1998 Itoh G06F 3/03543
345/163
5,870,081 A * 2/1999 Wu G06F 3/03543
345/163
5,917,473 A * 6/1999 Yeh G06F 3/03543
345/184
6,157,370 A * 12/2000 Kravtin G06F 3/03543
248/118.1
6,304,249 B1 * 10/2001 Derocher G06F 3/03543
248/118.1
6,396,478 B1 * 5/2002 Kravtin G06F 3/03543
248/118.1
6,429,851 B1 * 8/2002 Vaghefi G06F 3/03543
D14/402
10,592,008 B1 * 3/2020 Wang G06F 3/0346
11,119,586 B2 * 9/2021 Zarnowitz G06F 3/03543
11,520,417 B2 * 12/2022 Lu G06F 3/039
11,762,484 B1 * 9/2023 Lu G06F 3/03543
345/163

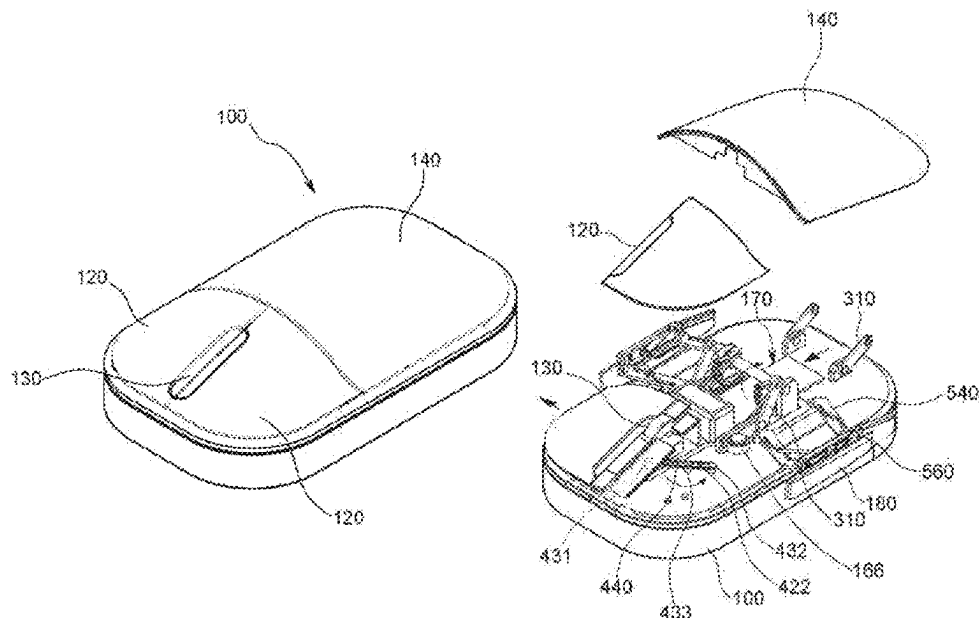
(Continued)

Primary Examiner — Benyam Ketema

(57) **ABSTRACT**

Disclosed is a computer mouse that may be changed in size and shape. According to an embodiment, the computer mouse includes a first driving unit moving a cover in the front and rear direction to perform an up-and-down operation, a second driving unit moving a plurality of buttons in the left and right direction to perform an open-and-close operation, a sub-button moving the cover backward to raise the cover and at the same time is drawn out to both sides of the body or is drawn into the body when the cover is moved forward, and a third driving unit causing the sub-button to appear and disappear. In addition, a computer mouse that is easy to carry is disclosed. According to an embodiment, the computer mouse includes a mouse body, and a cover member detachably coupled to the mouse body.

6 Claims, 17 Drawing Sheets



(56)	References Cited				2006/0033714	A1 *	2/2006	Boldin		G06F 3/03543		
	U.S. PATENT DOCUMENTS				2006/0170655	A1 *	8/2006	Hou		G06F 3/03543		
	2001/0005198	A1 *	6/2001	Sakamoto		G06F 3/03543	2006/0227109	A1 *	10/2006	Lo		G06F 3/03543
	2002/0190952	A1 *	12/2002	Shah		G06F 3/03543	2006/0290664	A1 *	12/2006	Sherman		G06F 3/03543
	2003/0025672	A1 *	2/2003	Collins		G06F 3/038	2007/0132733	A1 *	6/2007	Ram		G06F 3/03544
	2003/0103040	A1 *	6/2003	Koike		G06F 3/03543	2008/0170360	A1 *	7/2008	Cheng		G06F 3/03543
	2003/0214484	A1 *	11/2003	Haywood		G06F 3/0338	2009/0046064	A1 *	2/2009	Manalo		G06F 3/03543
	2004/0169640	A1 *	9/2004	Chao		G06F 3/03543	2017/0192536	A1 *	7/2017	Wang		G06F 3/03543
	2005/0052417	A1 *	3/2005	Chi		G06F 3/0231	2017/0300137	A1 *	10/2017	Chen		G06F 3/03543
	2005/0116934	A1 *	6/2005	Yin		G06F 3/03543	2019/0056805	A1 *	2/2019	Wang		G06F 3/0362
	2006/0007151	A1 *	1/2006	Ram		G06F 3/0383	2022/0011873	A1 *	1/2022	Kauh		G06F 3/0362
							2023/0376127	A1 *	11/2023	Helmes		G06F 3/039
							</					

* cited by examiner

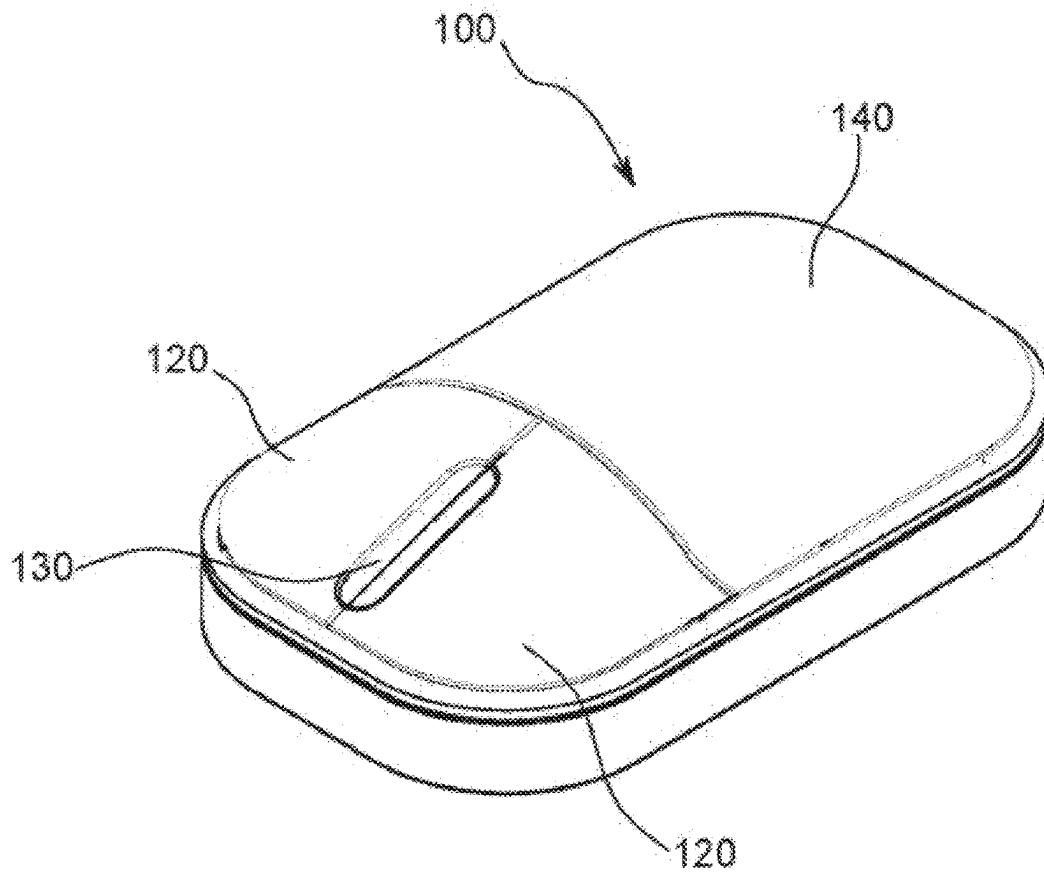


FIG. 1

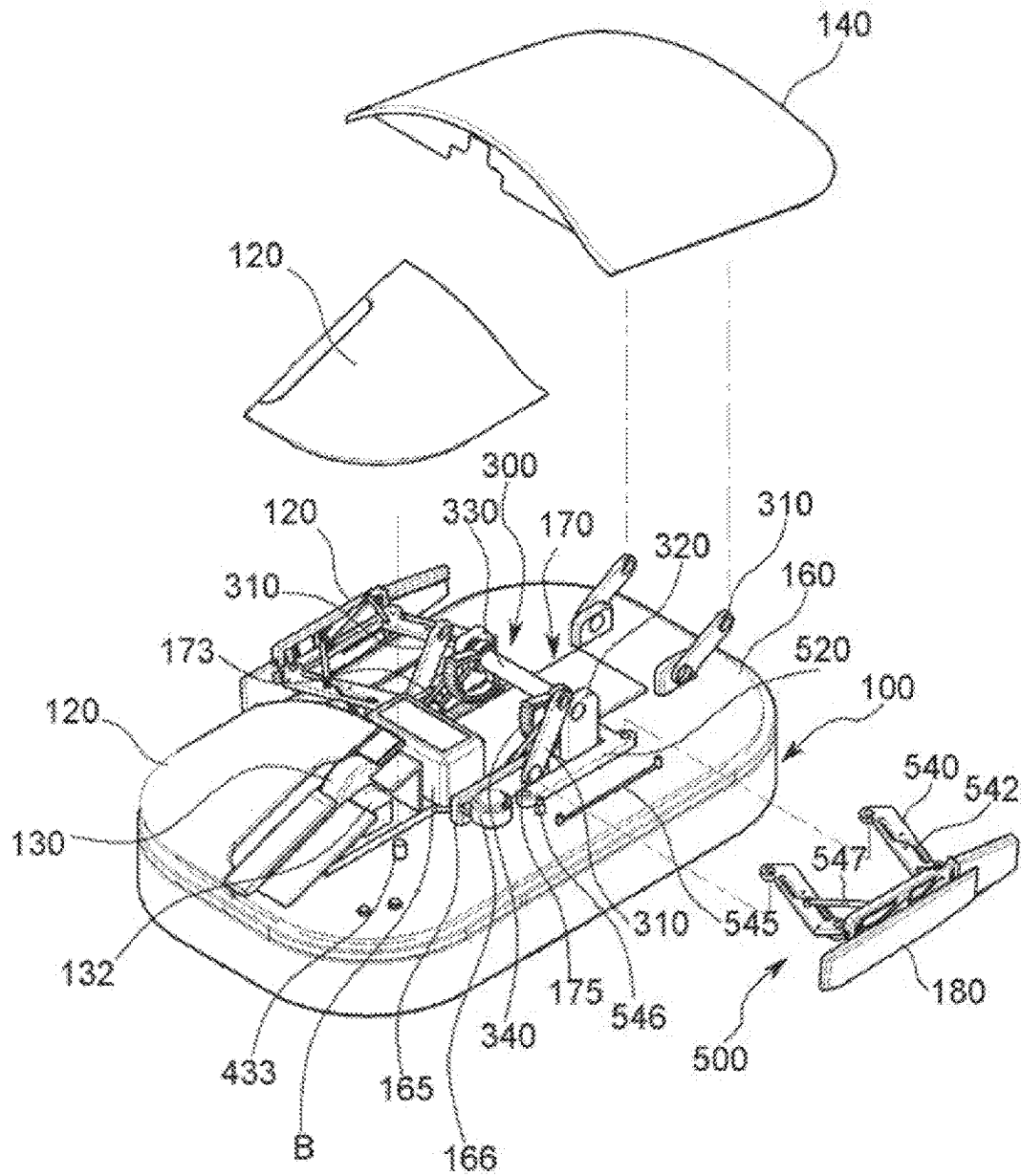


FIG. 2

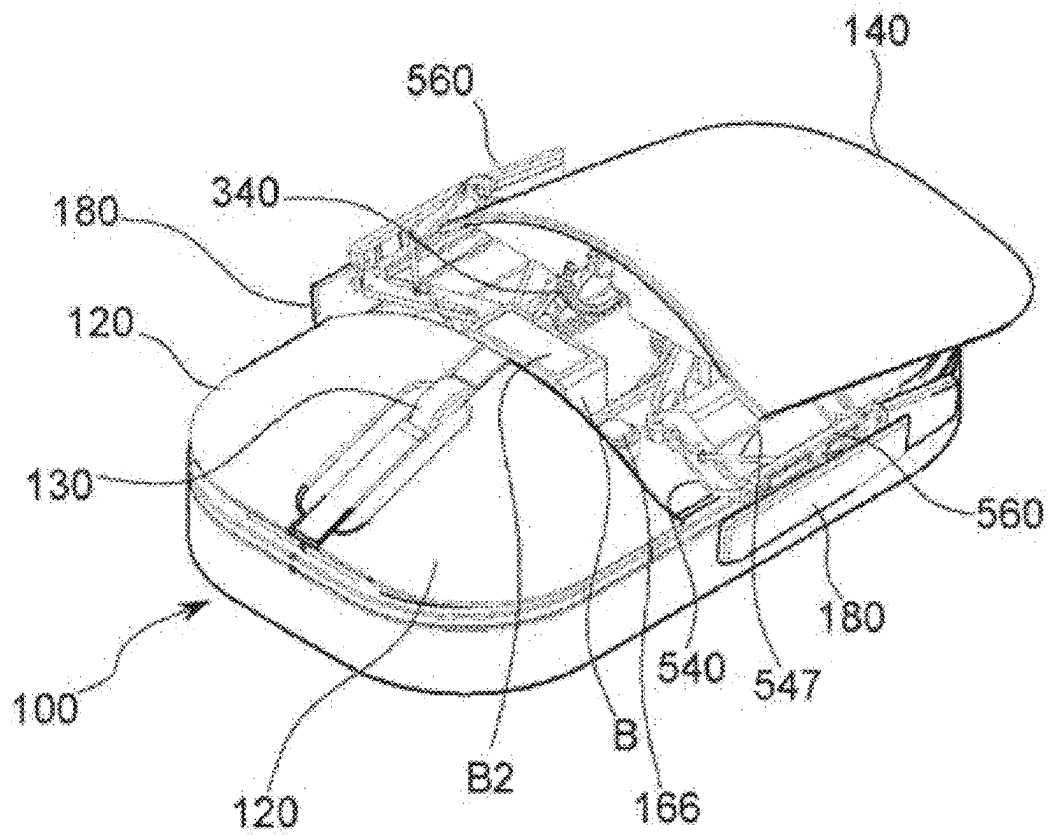


FIG. 3

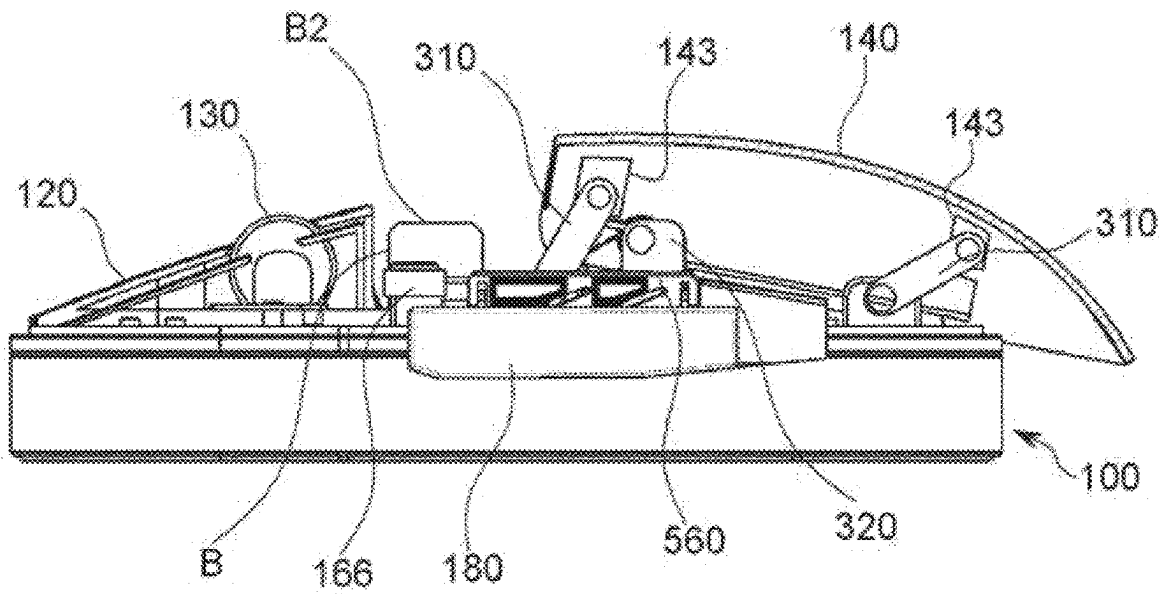


FIG. 4

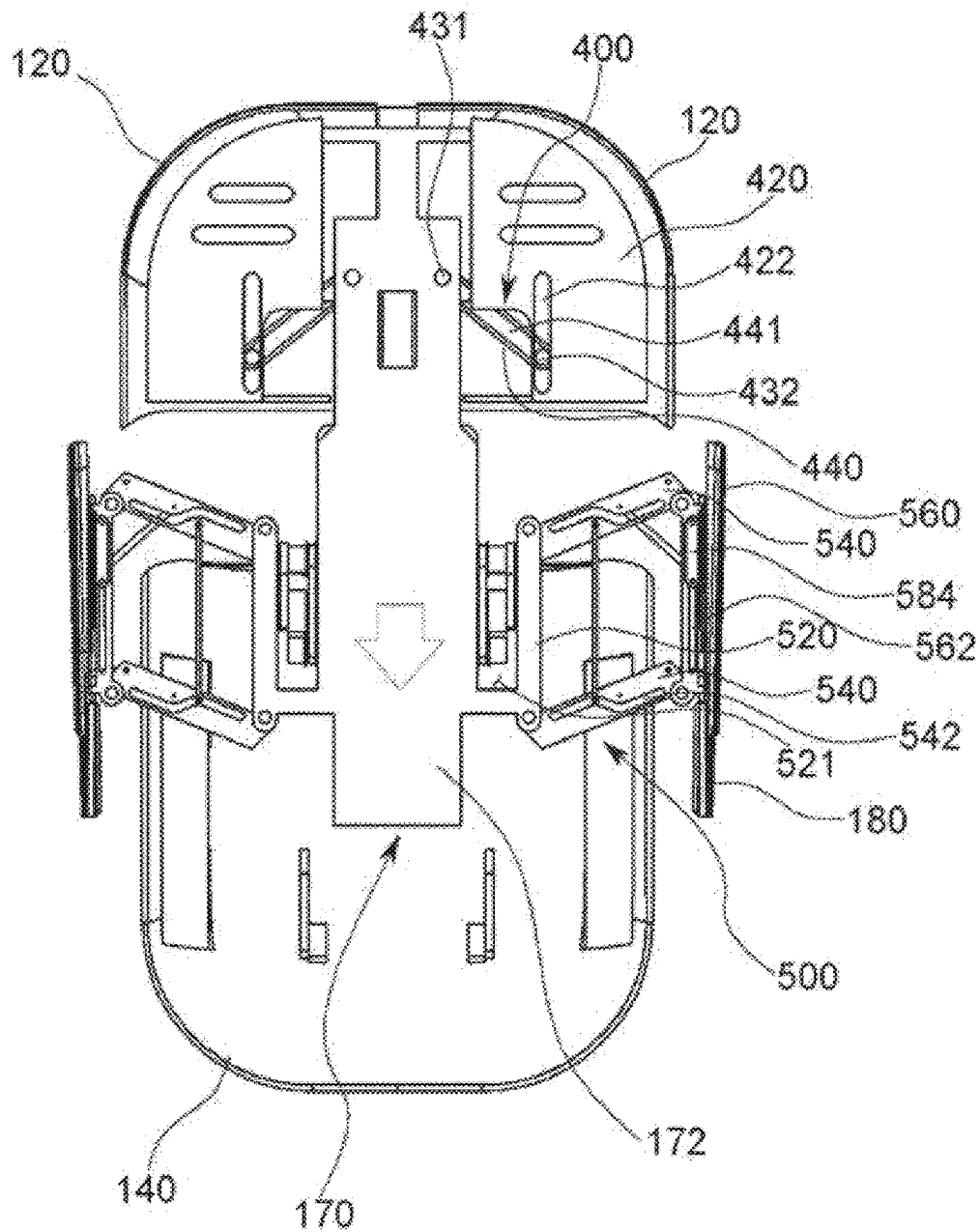


FIG. 5

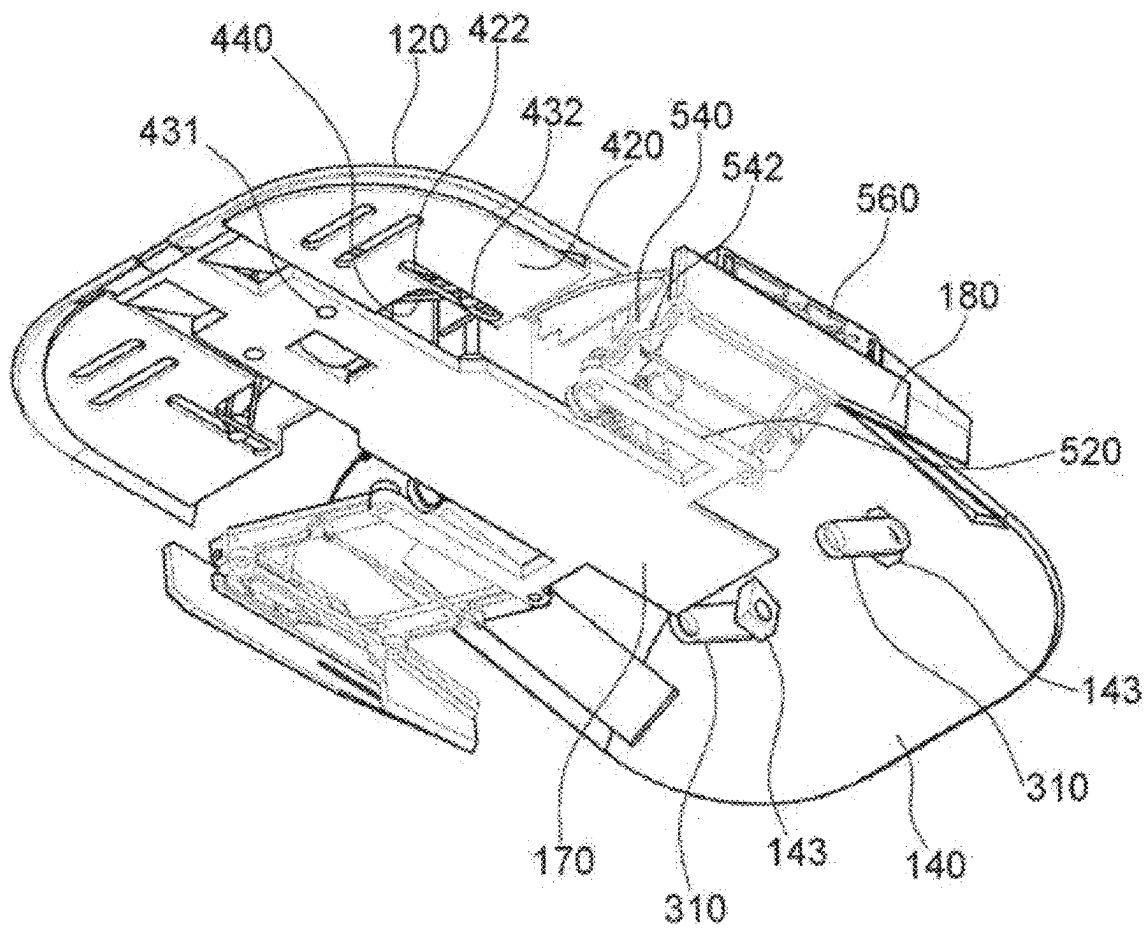


FIG. 6

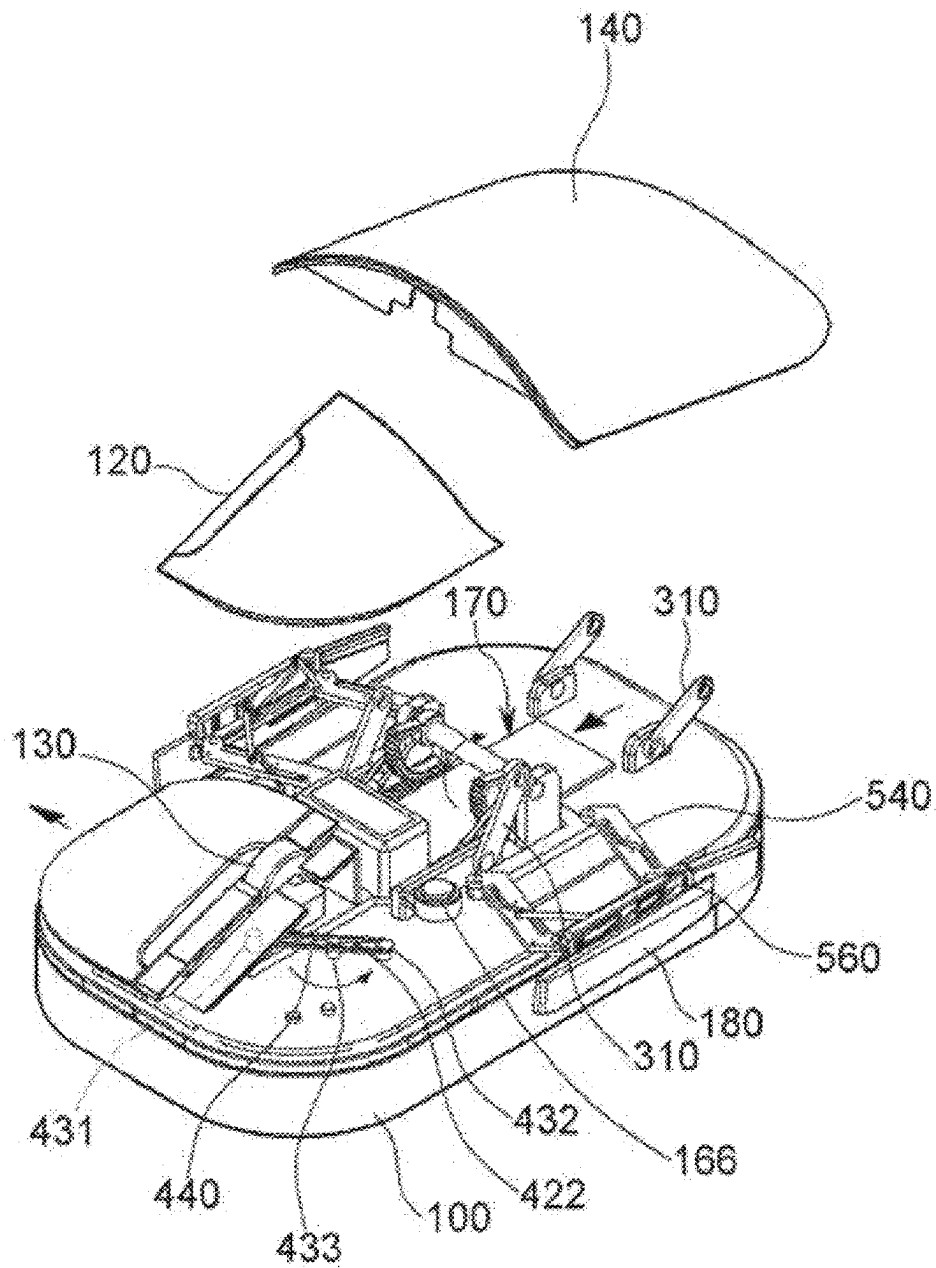


FIG. 7

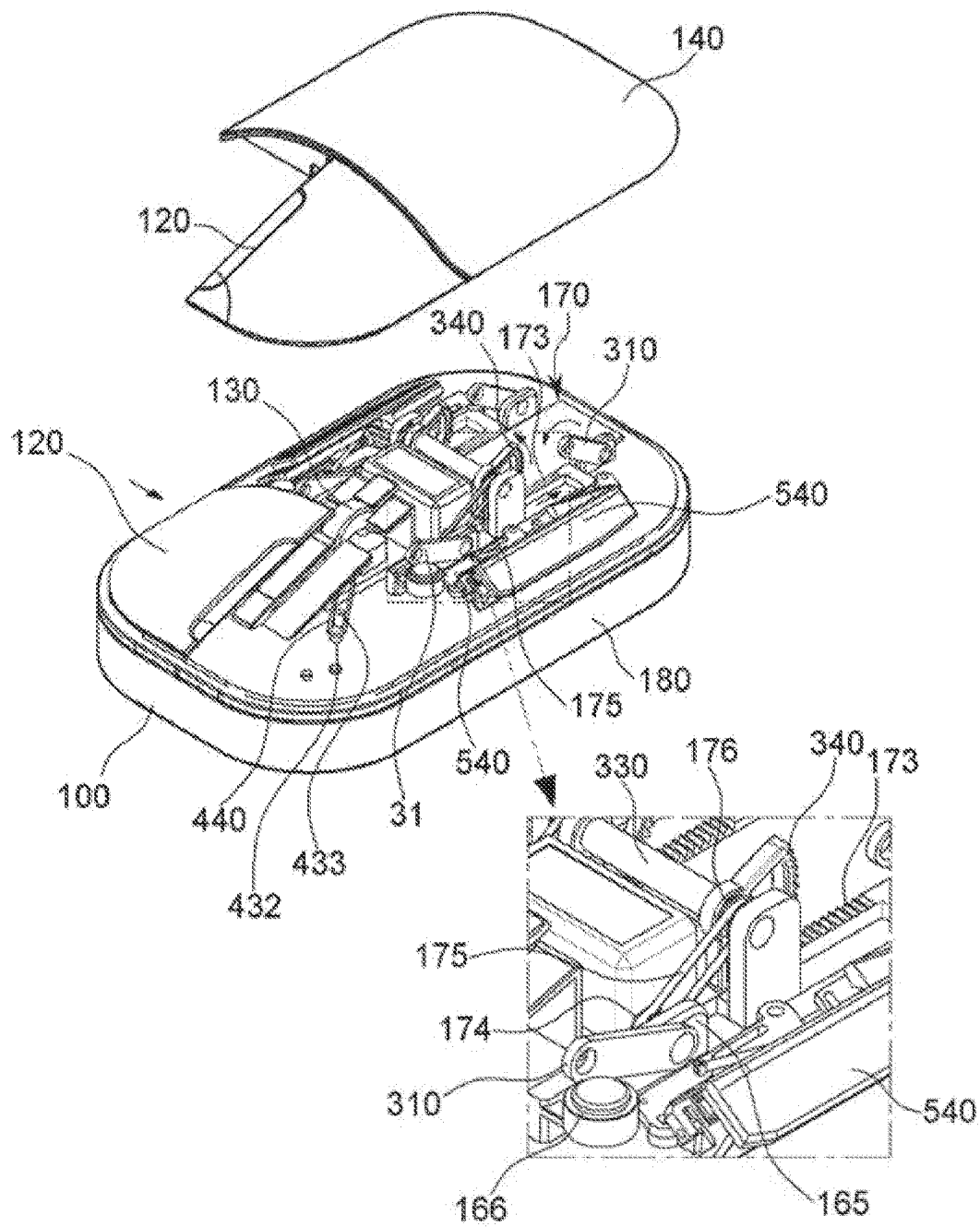


FIG. 8

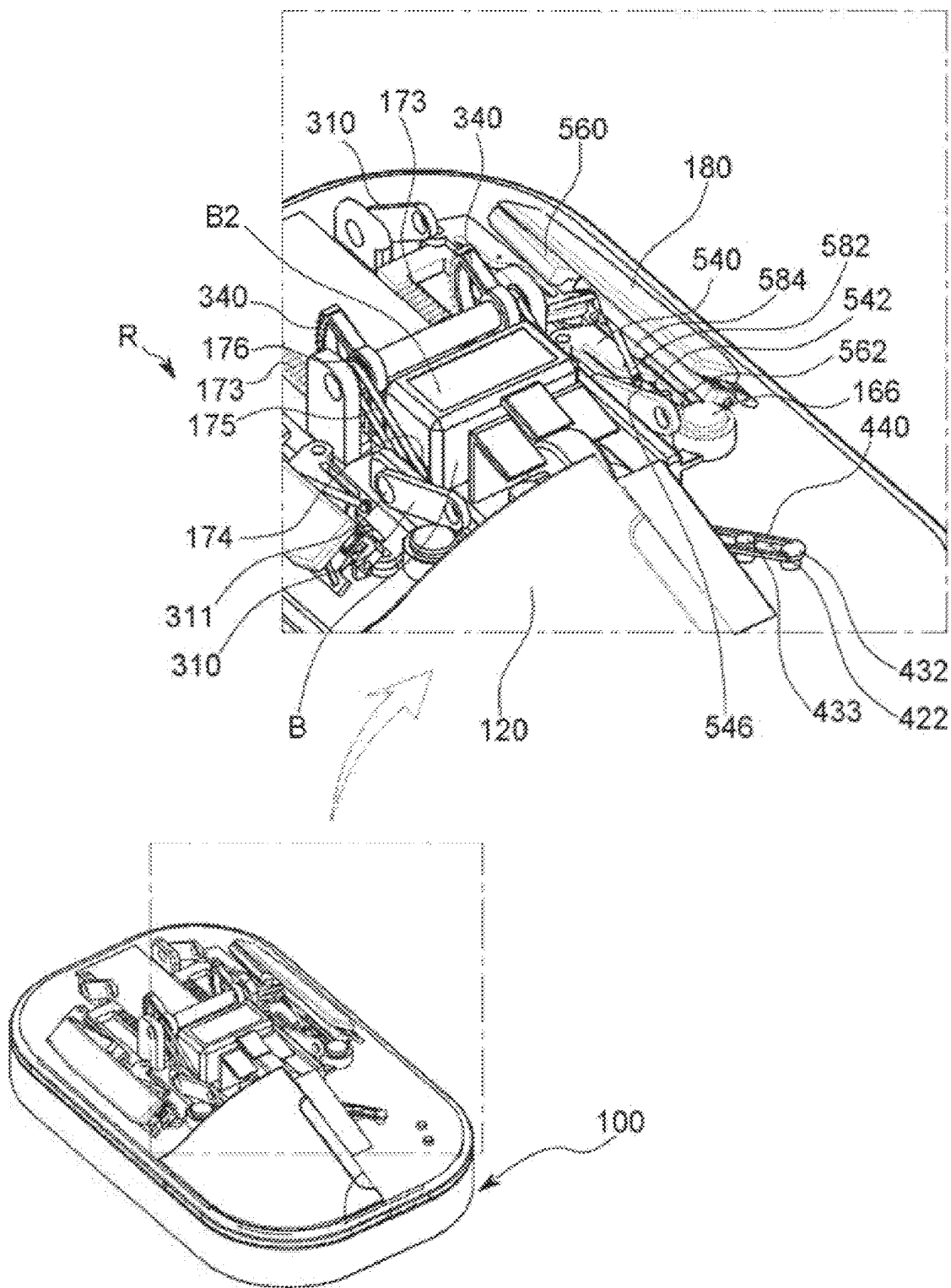


FIG. 9

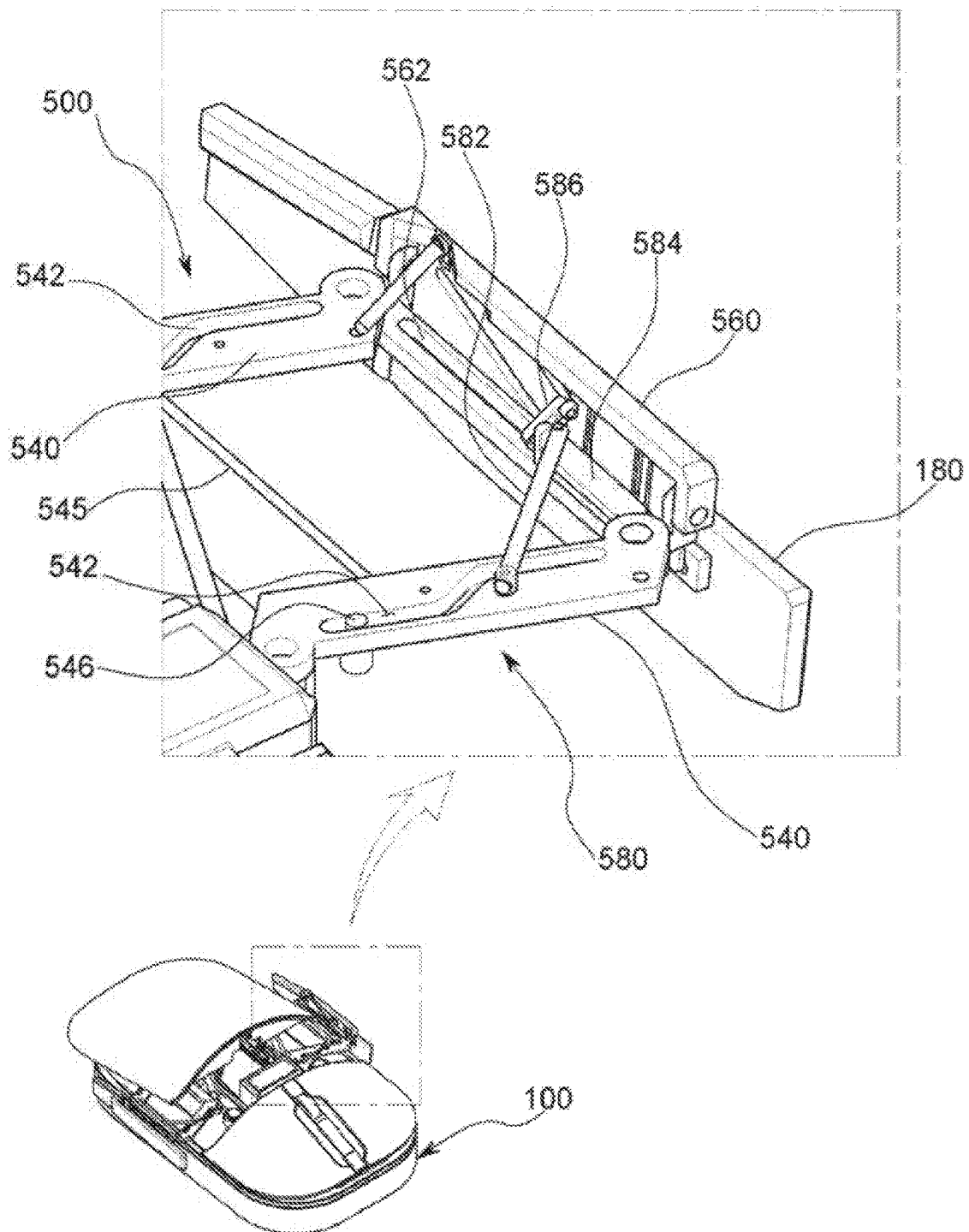


FIG. 10

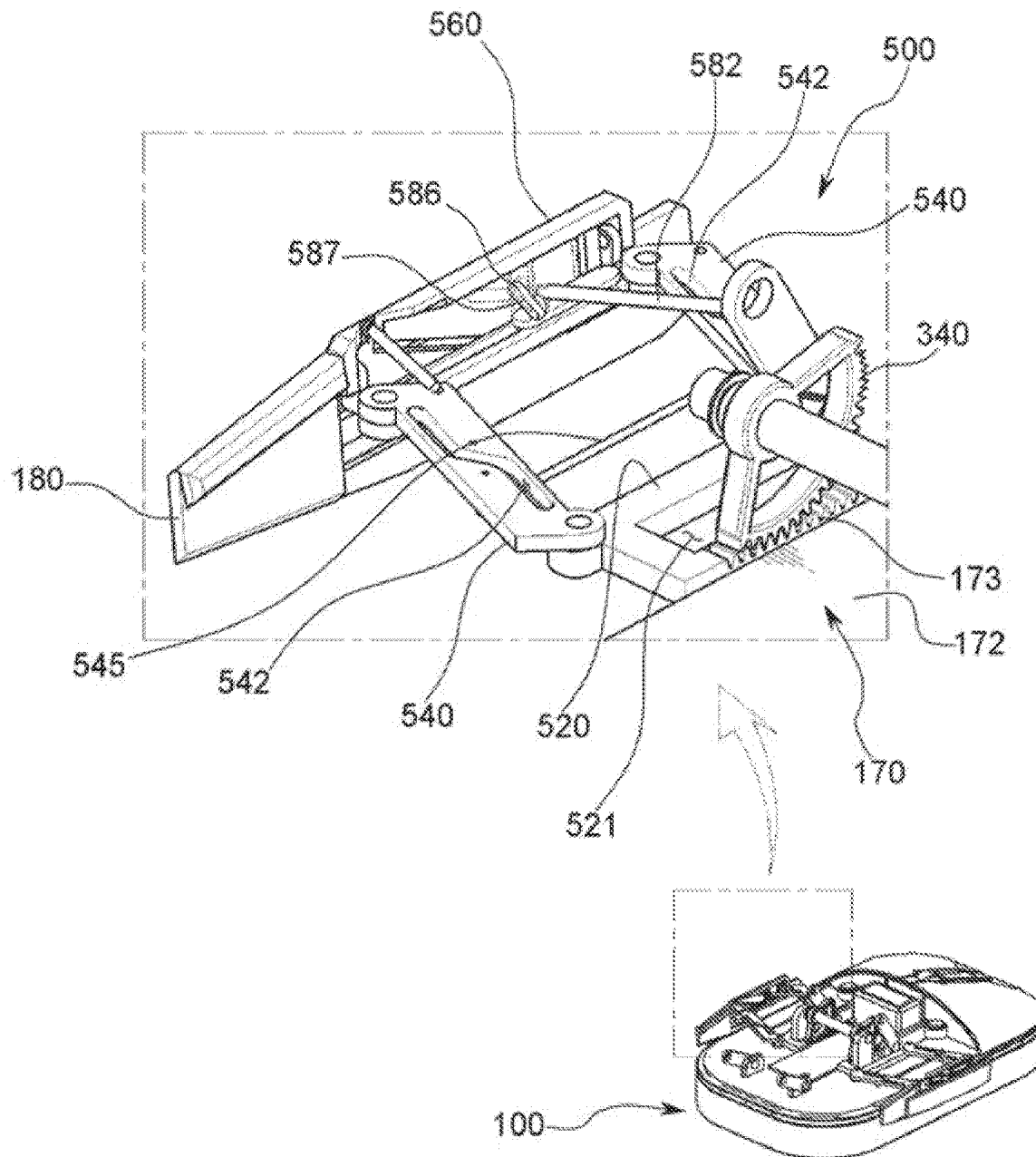


FIG. 11

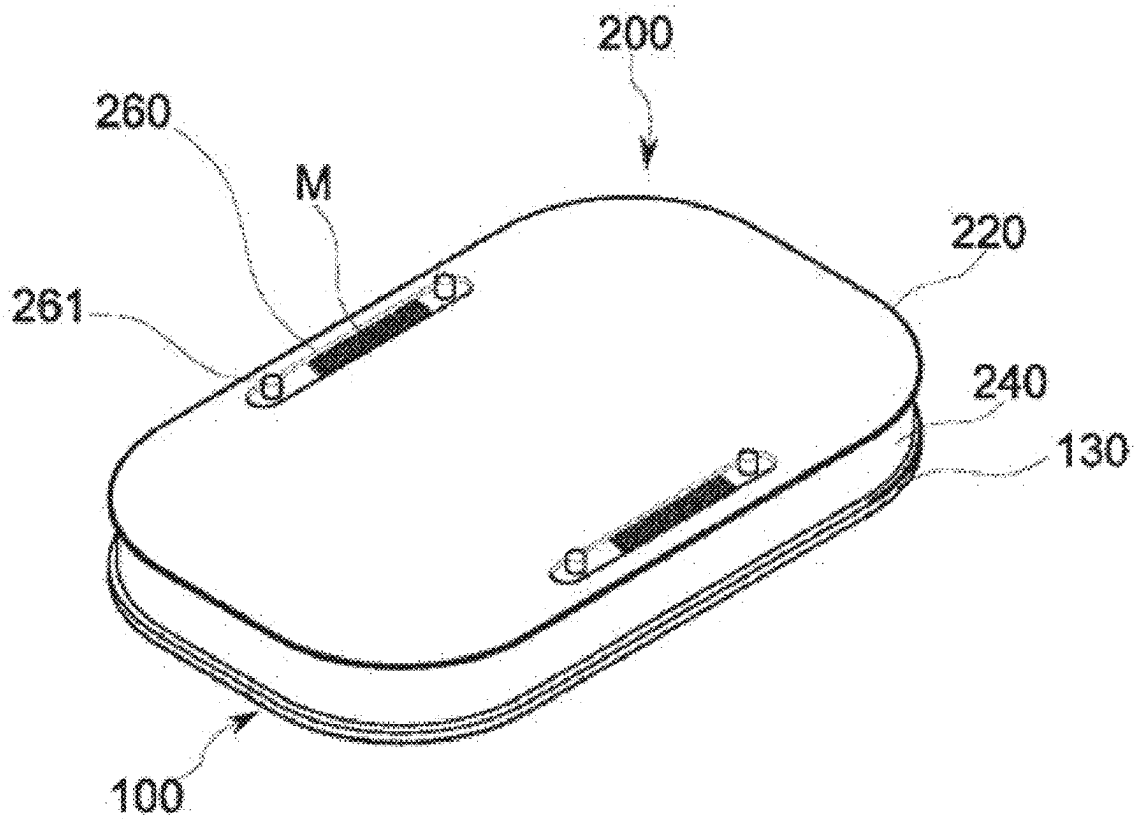


FIG. 12

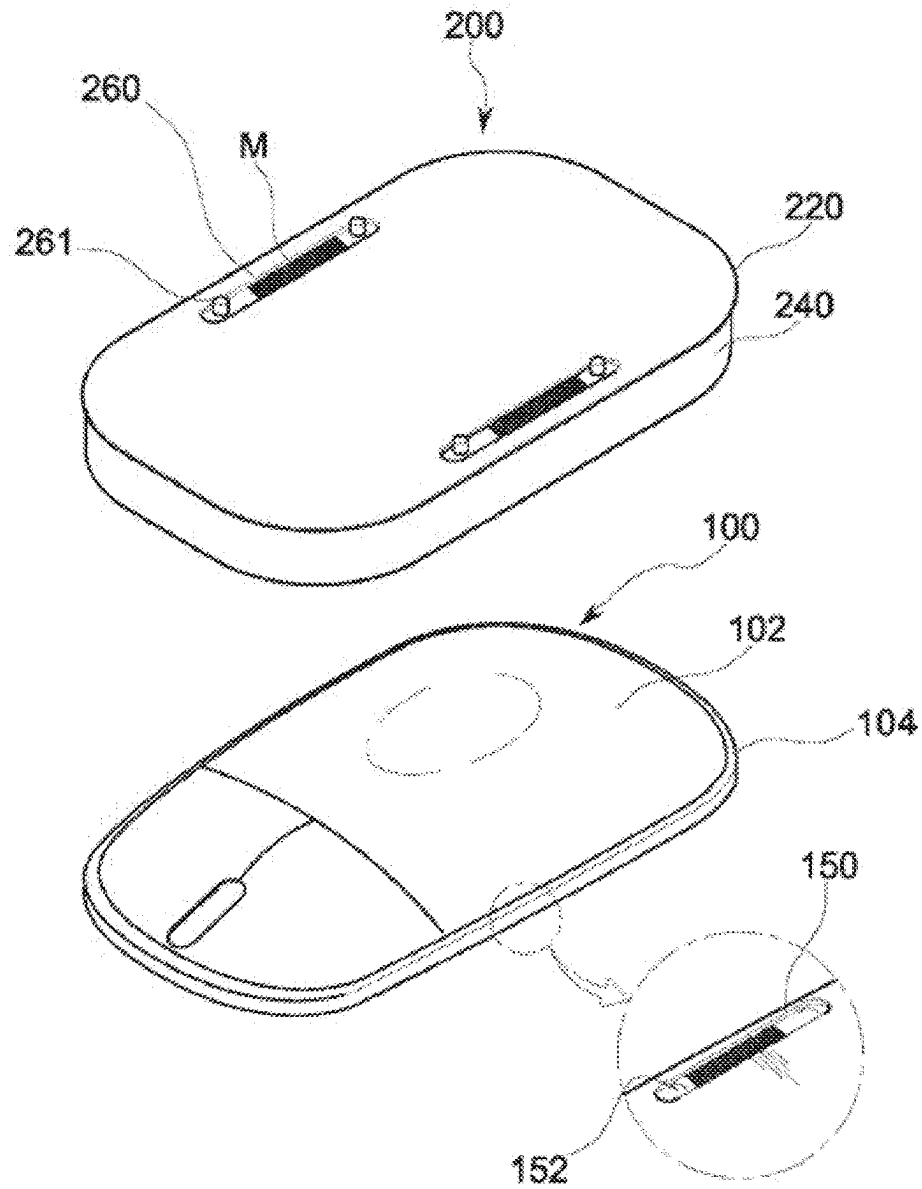


FIG. 13

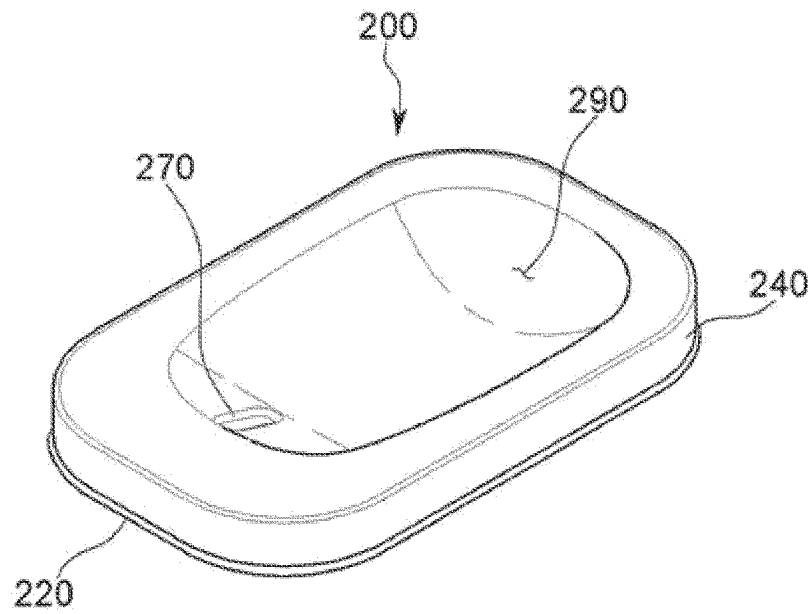


FIG. 14

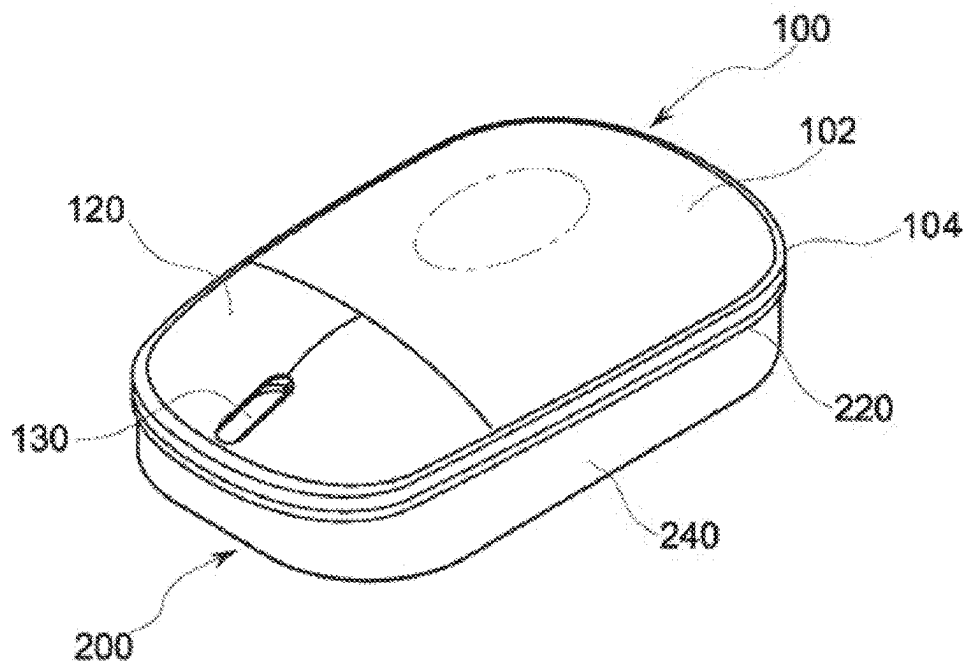


FIG. 15

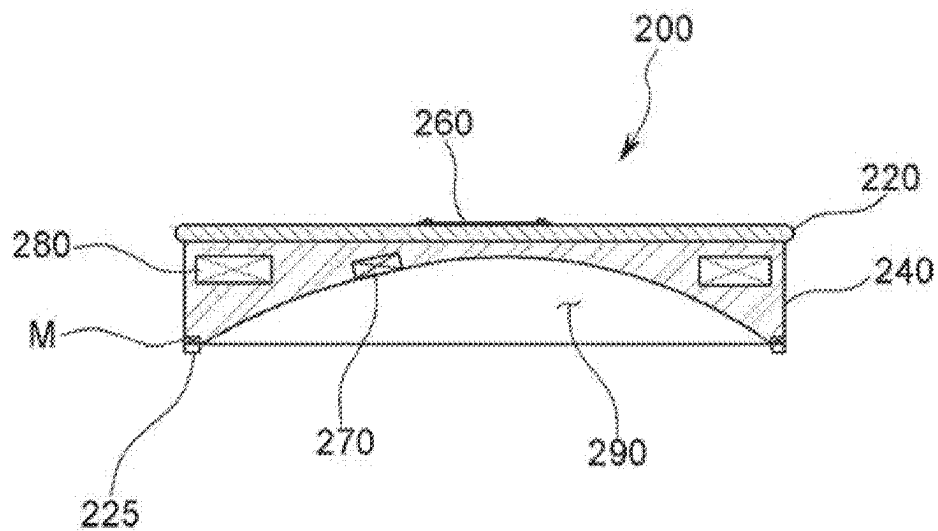


FIG. 16

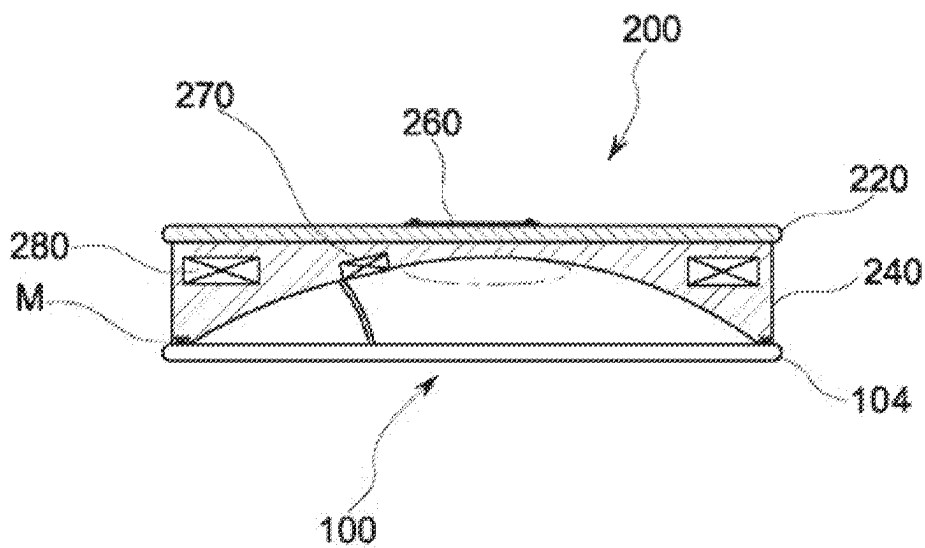


FIG. 17

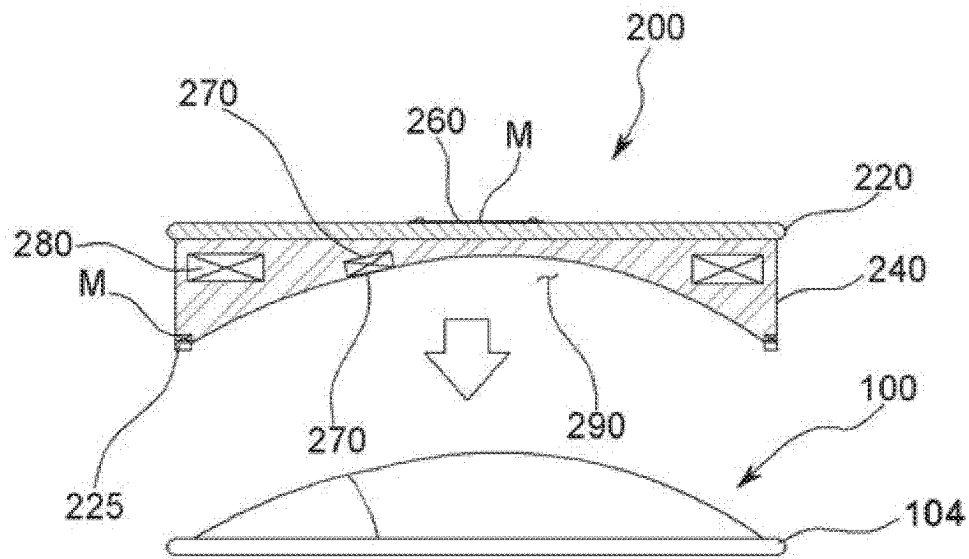


FIG. 18

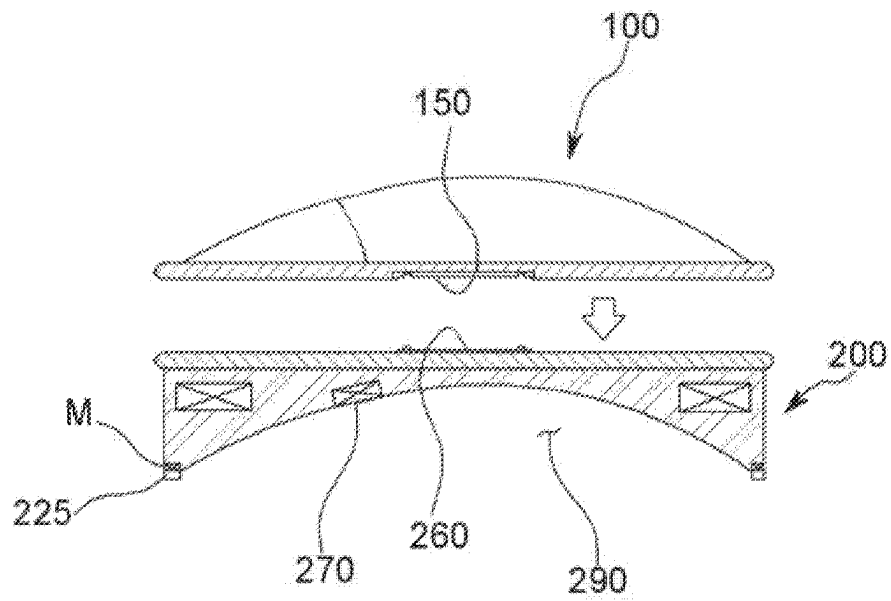


FIG. 19

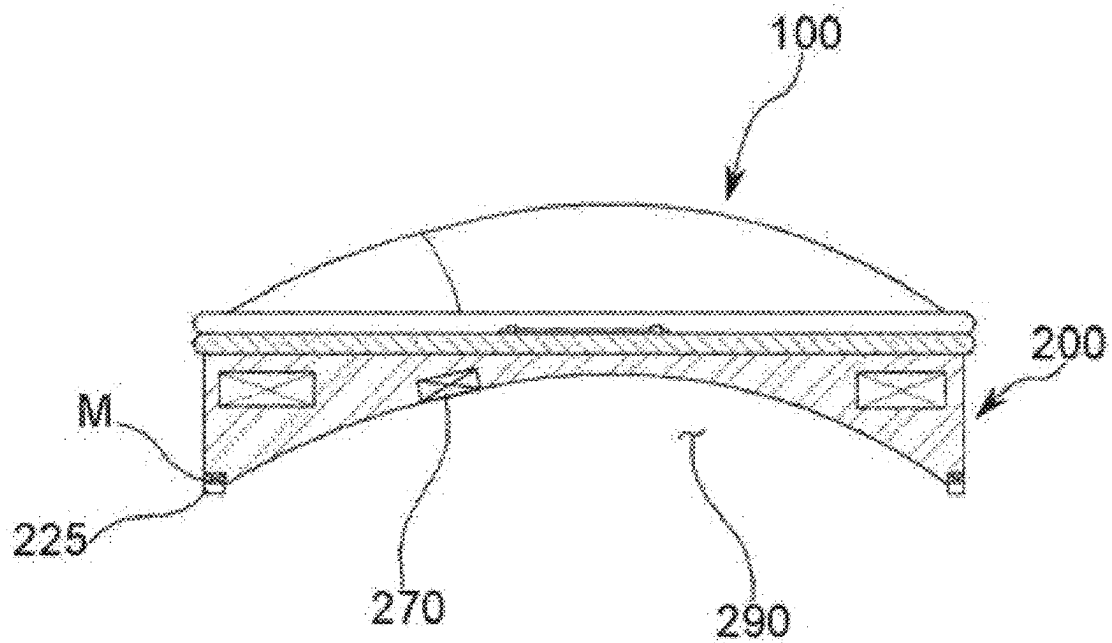


FIG. 20

1

COMPUTER MOUSE

TECHNICAL FIELD

The present invention relates to a computer mouse, and more particularly to, a computer mouse that is changed in size and shape to suit the purpose or preferred size of each individual, is easy to carry, and has a beautiful and luxurious appearance.

BACKGROUND ART

A mouse, which is one of input devices of a computer, is formed in a small size so as to fit in the hand while forming a convex shape that is easy to hold with the hand.

The mouse is connected to a computer body by wired or wirelessly, a plurality of buttons are formed to move (input) a cursor on a screen, and a wheel is provided between these buttons.

The mouse is formed convexly upward so as to be comfortable to hold while wrapping the hand around the mouse, and is used by putting the palm lightly on an upper surface of the mouse and by holding both sides with the thumb and little finger.

In general, since a computer mouse is constantly exposed to the outside, contamination is concerned and there is a risk of damage due to impact.

In addition, since the computer mouse according to the prior art is usually treated as an inexpensive consumable, it is usually manufactured of a low-cost material made of synthetic resin, and thus, there was a disadvantage in failing to satisfy the needs of consumers who want a luxurious image.

In addition, since the computer mouse according to the prior art has a fixed size, it was inconvenient for a person with a large hand to use it because a grip portion is small, and additional buttons that may more cover functions such as DPI and backspace were needed.

OBJECT OF THE INVENTION

It is an object of the present invention to provide a computer mouse that is used in a normal type size, may be used comfortably by a person with big hands by increasing the width and height of a grip portion if necessary, may be changed in the size and shape so as to be optimized for long-term detailed work, may operate noiselessly in a casual mode, and may operate noisily in a professional mode.

It is another object of the present invention to provide a computer mouse that is easy to carry and includes a cover made of a luxurious material to protect a button portion from external impact and to have a beautiful appearance, the cover being electrically connected to a body to enable signal transmission and reception and power supply.

Technical Solution

To accomplish the above-mentioned objects, according to an aspect of the present invention, there is provided a computer mouse including a mouse body having a plurality of buttons on an upper portion and a scroll wheel formed between the plurality of buttons, and formed in an upwardly convex shape, wherein the mouse body includes a support unit including a plate formed on a lower portion and a cover formed on an upper portion, having a space formed inside, and formed in the middle of the space, a first driving unit moving the cover in a front and rear direction to perform an

2

up-and-down operation, a second driving unit moving the plurality of buttons in a left and right direction to perform an open-and-close operation, a sub-button moving the cover backward to raise the cover and at the same time, drawn out to both sides of a body or drawn into the body when the cover is moved forward, and a third driving unit causing the sub-button to appear and disappear.

The support unit may include a plate member placed on an upper portion of the plate and having rack gears formed on both sides of an upper surface, and a box portion protruding from an upper portion of the plate member, a liquid crystal display unit may be formed on an upper surface of the box portion, a plurality of setting switches may be formed, and the plurality of setting switches may set functions of DPI, lighting color of an LED, and a sub-button.

A scroll wheel may be formed in front of the box portion.

The first driving unit may include an operation axis having an upper end hinged to a bottom surface of the cover and a lower end hinged to a protrusion formed on an upper surface of the plate; and a rotating portion including a boss protruding from the upper surface of the plate, a shaft coupled to the boss, a main gear coupled to the shaft, a rack gear formed on an upper surface of the support unit in a longitudinal direction so as to be geared to the main gear, a first pulley formed on a lower hinge axis of the operation axis, a second pulley formed on the shaft, and a belt connecting the first pulley to the second pulley.

The second driving unit may include a horizontal plate formed on a lower portion of the button and having a slide long hole formed; and a link having one end fitted to a first axis pin formed in front of the support unit, the other end fitted to a second axis pin inserted into the long slide hole of the horizontal plate, and having a long hole formed to guide driving of the first axis pin.

The third driving unit may include a connection bar formed on a side surface of the support unit and having a groove in which a boss and a protrusion of the plate are accommodated; a side link bar having one end hinged to the connection bar and having a slide groove formed to have a length; and a side bar hinged to the other end of the side link bar, and the sub-button may be coupled to the side bar.

The computer mouse according may include a folding unit rotating the sub-button to fold the sub-button in an up and down direction, and the folding unit may include a rod having one end inserted into the slide groove of the side link bar to be movable, and the other end hinged to the side bar; a slider coupled to a long hole formed in the side bar in the longitudinal direction and to which the other end of the rod is hinged; and an induction link having a long hole of which one end is fitted to an axis pin protruding from an upper surface of the slider, and the other end is hinged to the side bar.

To accomplish the above-mentioned objects, according to another aspect of the present invention, there is provided a computer mouse that is easy to carry including a mouse body and a cover member detachably coupled to the mouse body, wherein, in the cover member, a panel portion is formed on an upper portion, a side wall is formed on a lower portion of the panel portion, and an insertion groove into which an upper portion of the mouse body is inserted is formed concavely on a lower surface, a magnetic body is formed on the panel portion, an edge portion magnetically attached to the magnetic body is formed on an outer circumferential surface of the mouse body so that the cover member may be coupled to the edge portion by magnetic attachment, a first terminal unit is formed in the panel portion, a second

3

terminal unit in contact with the first terminal unit is formed in the lower portion of the mouse body so that a sensor signal and power are supplied.

In the cover member, a sensor may be formed in an inner circumferential surface of an insertion groove, a power supply unit may be formed inside, the sensor and the first terminal unit may be electrically connected so that a signal read from the sensor is transmitted to the first terminal unit.

The mouse body may include a plurality of buttons and a body mounted on an upper portion, formed in an arc shape, and made of stainless or aluminum, a touch-responsive wheel may be formed between the plurality of buttons, a convex protrusion may be formed in the first terminal unit, and the second terminal unit may be formed as a concave recessed portion.

Advantageous Effects of the Invention

According to the present invention, the computer mouse is used in the normal type size, may be used comfortably by a person with big hands by increasing the width and height of the grip portion if necessary, may be optimized for long-term detailed work through a shape change, may operate noiselessly in the casual mode, and may operate noisily in the professional mode, and thus, there is an effect that the deformation may be freely applied according to a place or a user's mood.

According to the present invention, the computer mouse is easy to carry and provides the cover made of a luxurious material to protect the mouse body and the button portion from external impact and the mouse body, and thus, there is an effect of having a beautiful appearance.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view showing a casual mode of a computer mouse according to the present invention.

FIG. 2 is an exploded perspective view showing a computer mouse according to the present invention.

FIG. 3 is a perspective view showing a pro mode of a computer mouse according to the present invention.

FIG. 4 is a side cross-sectional view showing a computer mouse according to the present invention.

FIG. 5 is a bottom view showing the inside of a computer mouse according to the present invention.

FIG. 6 is a bottom perspective view showing the inside of a computer mouse according to the present invention.

FIG. 7 is a perspective view showing the inside of a computer mouse in a pro mode according to the present invention.

FIG. 8 is a perspective view showing the inside of a computer mouse in a casual mode according to the present invention.

FIG. 9 is a perspective view showing a folded state of a 'third driving unit' of a computer mouse according to the present invention.

FIG. 10 and FIG. 11 are perspective views showing a deployed state of a 'third driving unit' of a computer mouse according to the present invention.

FIG. 12 is a perspective view showing a portable mode of a computer mouse according to the present invention.

FIG. 13 is an exploded perspective view of a computer mouse according to the present invention.

FIG. 14 is a perspective view of an inverted 'cover member' of a computer mouse according to the present invention.

4

FIG. 15 is a cross-sectional view showing a casual mode in which a cover member of a computer mouse is coupled to a lower portion of a mouse body according to the present invention.

FIG. 16 is a cross-sectional view of a 'cover member' of a computer mouse according to the present invention.

FIGS. 17 to 20 are diagrams sequentially showing modified examples of a computer mouse according to the present invention.

EXPLANATION OF REFERENCE NUMERALS

100: mouse body **120:** button
130: scroll wheel **140:** cover
150: second terminal unit **160:** plate
170: support unit **173:** rack gear
174: first pulley **176:** second pulley
175: belt **180:** sub-button
200: cover member **220:** panel portion
240: side wall **290:** insertion groove
225: pad M: magnetic body
260: first terminal unit **270:** sensor
280: power supply unit **300:** first driving unit
310: operation axis **320:** boss
330: shaft **340:** main gear
400: second driving unit **420:** horizontal plate
440: link **500:** third driving unit
520: connection bar **540:** side link bar
560: side bar **580:** folding unit
582: rod **584:** slider
586: induction link

DESCRIPTION OF THE INVENTION

Hereinafter, embodiments will be described with reference to the accompanying drawings. However, various changes may be made to the embodiments, and thus, the scope of the patent application is not limited or restricted by these embodiments. It should be understood that all changes, equivalents or substitutes to the embodiments are included within the scope of rights.

Among the accompanied drawings, FIG. 1 is a perspective view showing a casual mode of a computer mouse according to the present invention, FIG. 2 is an exploded perspective view showing the computer mouse according to the present invention, FIG. 3 is a perspective view showing a pro mode of the computer mouse according to the present invention, FIG. 4 is a side cross-sectional view showing the computer mouse according to the present invention, FIG. 5 is a bottom view showing the inside of the computer mouse according to the present invention, FIG. 6 is a bottom perspective view showing the inside of the computer mouse according to the present invention, FIG. 7 is a perspective view showing the inside of the computer mouse in a pro mode according to the present invention, FIG. 8 is a perspective view showing the inside of the computer mouse in a casual mode according to the present invention, FIG. 9 is a perspective view showing a folded state of a 'third driving unit' of the computer mouse according to the present invention, FIG. 10 is a perspective view showing a deployed state of a 'third driving unit' of the computer mouse according to the present invention.

The computer mouse according to the present invention includes a mouse body **100** having a plurality of buttons **120** on an upper portion and a scroll wheel **130** formed between the plurality of buttons **120**, and formed in an upwardly

5

convex shape. The scroll wheel **130** may be a touch type scroll wheel. This state is referred to as a “casual mode”.

In the casual mode, the front button **120** is implemented in a noiseless manner. The casual mode is applied when using in a relatively quiet place such as an office.

On the other hand, when changed to a “pro mode” to be described below, the button **120** may be implemented in a noisy manner. The pro mode is used when a relatively active movement is required, such as in a game, and may give a cheerful feeling by generating a clicking sound.

The mouse body **100** includes a support unit **170** including a plate **160** formed on a lower portion and a cover **140** formed on an upper portion, having a space formed inside, and formed in the middle of the space,

a first driving unit **300** moving the cover **140** in the front and rear direction to perform an up-and-down operation,

a second driving unit **400** moving the plurality of buttons **120** in the left and right direction to perform an open-and-close operation,

a sub-button **180** moving the cover **140** backward to raise the cover **140** and at the same time, drawn out to both sides of a body **102** or drawn into the body **102** when the cover **140** is moved forward; and a third driving unit **500** causing the sub-button **180** to appear and disappear.

As for the reference to the direction described in the present invention, it should be noted that forward means the left side, and backward means the right side with reference to FIG. 1.

The support unit **170** includes a plate member **172** placed on an upper portion of the plate **160** and having rack gears **173** formed on both sides of an upper surface, and a box portion **B** protruding from an upper portion of the plate member **172**.

A liquid crystal display unit **B2** is formed on an upper surface of the box portion **B**, and the liquid crystal display unit **B2** displays numerical values digitally.

A plurality of setting switches **166** are formed on one side of the liquid crystal display unit **B2** and on the upper portion of the plate **172**.

By manipulating the plurality of setting switches **166**, functions of DPI, lighting, and sub-buttons are set.

In addition, the scroll wheel **130** is provided in front of the box portion **B**. Since the scroll wheel **130** is mechanically driven, it is coupled between the brackets **132** formed on both sides of the upper portion of the plate member **172** to be rotatable.

In addition, an LED (not shown) is configured on the upper portion of the plate **160** or the support unit **170** and is turned on when the cover **140** is opened, so that a lighting function may be added.

The first drive unit **300** performing an open-and-close operation on the cover **140** includes an operation axis **310** having an upper end hinged to a bottom surface of the cover **140** and a lower end hinged to a protrusion **165** formed on the upper surface of the plate **160**; and a rotating portion **R** including a boss **320** protruding from the upper surface of the plate **160**, a shaft **330** coupled to the boss **320**, a main gear **340** coupled to the shaft **330**, a rack gear **173** formed on the upper surface of the support unit **170** in a longitudinal direction so as to be geared to the main gear **340**, a first pulley **174** formed on a lower hinge axis **311** of the operation axis **310**, a second pulley **176** formed on the shaft **330**, and a belt **175** connecting the first pulley **174** to the second pulley **176**.

6

The operating axes **310** are provided at the front and rear, hinged to the upper surface of the plate **160**, provided in pairs on both sides, and coupled to a hinge connection portion **143** formed on a bottom surface of the cover **140** with a pin (FIG. 2 and see FIG. 4).

The main gear **340** is formed in a fan shape to have an arc portion, has teeth formed in the arc portion to be geared to the rack gear **173**, and has the other side coupled to the shaft **330** to be rotated integrally with the shaft **330**.

Therefore, as shown in FIG. 7 or 8, when the operation axis **310** is moved forward and backward by the front and rear movement of the cover **140**, since the first pulley **174** connected thereto is rotated, the belt **175** is rotated, and the second pulley **176** is rotated, the main gear **340** is rotated up and down, and the rack gear **173** geared to the main gear **340** is pushed in the front and rear direction, and thus, the support unit **170** may be moved in the front and rear direction.

Preferably, as shown in FIG. 4, when the cover **140** pulls backward and is opened, the cover **140** may be opened backward and rise by placing the left operation axis **310** at a higher position than the right operation axis **310**, and thus, the height may be adjusted upward.

Meanwhile, the second driving unit **400** performing an open-and-close operation on the button **120** in the left and right direction will be described.

As shown in FIGS. 5 to 8, the second driving unit **400** includes a horizontal plate **420** formed on the lower portion of the button **120** and having a slide long hole **422** formed; and a link **440** having one end fitted to a first axis pin **431** formed in front of the support unit **170**, the other end fitted to a second axis pin **432** inserted into the long slide hole **422** of the horizontal plate **420**, and having a long hole **441** formed to guide the driving of the first axis pin **431**.

Referring to FIG. 7, a reference pin **433** is coupled to the center of the link **440**. The reference pin **433** is fixed to the plate **160**, and thus, there is no change in the position.

Therefore, the link **440** rotates in the front and rear direction around the reference pin **433**.

Upon describing the operation of the second driving unit **400**, as shown in FIG. 8, when the support unit **170** moves backward, the first axis pin **431** is moved backward and rotates the link **440**, and the second axis pin **432** connected to the link **440** is moved along the slide long hole **422** of the link **440**, and thus, the second driving unit **400** operates in a direction in which the button **120** is closed.

The direction in which the button **120** is closed means an operation in which the buttons **120** on both sides are gathered together.

Conversely, as shown in FIG. 7, when the support unit **170** moves forward, the first axis pin **431** is moved forward and rotates the link **440** in a reverse direction, and the second axis pin **432** connected to the link **440** is moved in the reverse direction along the slide long hole **422** of the link **440**, and thus, the second driving unit **400** operates in a direction in which the buttons **120** on both sides are opened.

Opening means an operation in which the buttons **120** on both sides are opened in opposite directions to be apart from each other so that the scroll wheel **130** is exposed.

Meanwhile, the third drive unit **500** allowing the sub-button **180** to appear and disappear will be described with reference to FIGS. 5 and 9 to 11.

The third driving unit **500** includes a connection bar **520** formed on a side surface of the support unit **170** and having a groove **521** in which the boss **320** and the protrusion **165** of the plate **160** are accommodated; a side link bar **540** having one end hinged to the connection bar **520** and having a slide groove **542** formed to have a length; a side bar **560**

7

hinged to the other end of the side link bar **540**, and the sub-button **180** is coupled to the side bar **560**.

A protrusion **546** protruding from the upper surface of the plate **160** is coupled to the slide groove **542** of the side link bar **540**.

Referring to FIGS. **9** and **10**, while the side link bar **540** rotates along the protrusion **546**, the protrusion **546** may be guided to the slide groove **542**.

In addition, a link bar connecting rod **545** connecting the two side link bars **540** on both sides is provided, and thus, the two side link bars **540** may be maintained at a uniform interval without being opened or gathered each other during a rotation process.

The third driving unit **500** also includes a folding unit **580** rotating the sub-button **180** to perform a folding operation in the up and down direction.

The folding unit **580** includes a rod **582** having one end inserted into the slide groove **542** of the side link bar **540** to be movable, and the other end hinged to the side bar **560**; a slider **584** coupled to a long hole **562** formed in the side bar **560** in the longitudinal direction and to which the other end of the rod **582** is hinged; and an induction link **586** having a long hole **587** of which one end is fitted to an axis pin protruding from an upper surface of the slider **584**, and the other end is hinged to the side bar **560**.

As shown in FIGS. **3**, **10** and **11**, when the side link bar **540** is deployed, the rod **582** moves along the slide groove **542**, and the slider **584** connected to the other end of the rod **582** moves to the right along the long hole **562** of the sidebar **560**, and the axis pin of the slider **584** moves along the long hole **587** of the induction link **586** connected to the slider **584**, and thus, the sidebar **560** is erected vertically, and the sub-button **180** may be vertically disposed.

Therefore, as shown in FIGS. **3** and **5**, when it is necessary to perform an active operation such as a game, a mode is switched to a 'pro mode'.

That is, the cover **140** rises while moving backward, and the sub-buttons **180** are drawn out on the left and right sides respectively. When the mode is switched to the pro mode as described above, a click sound may be generated, and thus, the computer mouse is implemented noisily.

In addition, as the button **120** is opened to both sides, the scroll wheel **130** inside the button **120** is exposed.

The liquid crystal display unit **B2** of the box portion **B** is exposed.

In addition, it is possible to set DPI, on-off of the LED, color, and manipulation functions of the sub-button **180**, by manipulating the setting switch **166**.

When the sub-button **180** is drawn out, since the computer mouse is operated noisily, a user may feel clicking, and thus, a sense of satisfaction may be given to the user.

Meanwhile, a computer mouse further formed with a cover member so that the above-described computer mouse may be easy to carry and have a beautiful and luxurious appearance will be described. Here, FIGS. **12** to **20** briefly show the computer mouse shown in FIGS. **1** to **11** so that the main technical idea of the present invention may be clearly revealed.

Among the accompanied drawings, FIG. **12** is a perspective view showing a portable mode of a computer mouse according to the present invention, FIG. **13** is an exploded perspective view of a computer mouse according to the present invention, FIG. **14** is a perspective view of an inverted 'cover member' of a computer mouse according to the present invention, FIG. **15** is a cross-sectional view showing a casual mode in which a cover member of a computer mouse is coupled to a lower portion of a mouse

8

body according to the present invention, FIG. **16** is a cross-sectional view of a 'cover member' of a computer mouse according to the present invention, and FIGS. **17** to **20** are diagrams sequentially showing modified examples of a computer mouse according to the present invention.

According to the disclosed embodiment, the computer mouse includes a mouse body **100** and a cover member **200** detachably coupled to the mouse body **100**.

As shown in FIG. **12**, the computer mouse according to the present invention is convenient to carry and includes the mouse body **100** accommodated inside the cover member **200** to protect the mouse body **100** therein, which is referred to as a 'portable mode'.

In the cover member **200**, a panel portion **220** in the shape of a flat panel is formed on an upper portion, a side wall **240** is formed on a lower portion along an outer circumferential surface of the panel portion **220**, and an insertion groove **290** into which an upper portion of the mouse body **100** is inserted is formed concavely on a lower surface.

The panel portion **220** may be made of a synthetic resin material coated with leather or silicone to improve the quality of the material and improve grip feeling.

Preferably, at least one or more pads **225** may be formed on a lower end of the panel portion **220**. The pad **225** allows a lower surface of the cover member **200** to be apart from the ground and prevents excessive slipping, and is made of a Teflon material.

It will not be necessary to be limited to only Teflon material, and it will be natural that the pad **225** may be made of rubber or synthetic resin having elasticity.

Preferably, a mutual coupling means may be included so that the mouse body **100** may be fixed to the panel portion **220** of the cover member **200**.

As the coupling means according to an embodiment, a magnetic body **M** is formed on the panel portion **220** and the side wall **240**, and an edge portion **104** made of an iron (Fe) material that is magnetically attached to the magnetic body **M** is formed on the outer circumferential surface of the mouse body **100** so that the cover member **200** may be coupled to edge portion **104** by magnetic attachment.

A magnetic body (not shown) having a polarity different from that of the magnetic body **M** of the panel portion **220** may be formed on the edge portion **104**.

In the disclosed embodiment, a magnetic body is presented as the coupling means, but it is not necessarily limited thereto, and any means capable of connecting and fixing the cover member **200** to the mouse body **100** may be considered to be included in the concept of the present invention.

On the other hand, an electric conduction means is included to enable electrical signal transmission and power supply between the panel portion **220** and the mouse body **100**.

In the electric conduction means according to an embodiment, a first terminal unit **260** is formed in the panel portion **220**, and a second terminal unit **150** in contact with the first terminal unit **260** is formed in the lower portion of the mouse body **100** so that a sensor signal and power are supplied.

Referring to FIGS. **14** and **16**, in the cover member **200**, a sensor **270** is formed in an inner circumferential surface of an insertion groove **290**, a power supply unit **280** is formed inside, and the sensor **270** and the first terminal unit **260** are electrically connected so that a signal read from the sensor **270** is transmitted to the first terminal unit **260**.

It is preferable that an optical sensor is applied to the sensor **270** and a battery is applied to the power supply unit **280**.

Therefore, a position change signal detected by the sensor 270 may be transmitted to the first terminal unit 260, and may be electrically connected through the second terminal unit 150 connected to the first terminal unit 260 and formed in the mouse body 100 so that manipulation signals of the button 120 and the scroll wheel 130 may be transmitted to a computer.

The mouse body 100 includes the plurality of buttons 120 and the body 102 mounted on an upper portion and formed in an arc shape, made of stainless or aluminum, and the touch-responsive scroll wheel 130 is formed between the plurality of buttons 120.

Preferably, a convex protrusion 261 is formed in the first terminal unit 260 and the second terminal unit 150 is formed in a concave recessed portion 152 so that the first terminal unit 260 and the second terminal unit 150 are fitted each other, and thus, a contact state may be stably maintained.

The operation of the present invention configured as described above is described as follows.

When used in the portable mode, as shown in FIGS. 12 and 17, the cover member 200 is coupled from the upper portion of the mouse body 100 so that the mouse body 100 is inserted into the insertion groove 290.

When coupling is completed, the panel portion 220 of the cover member 200 and the edge portion 104 of the mouse body 100 are coupled by the coupling means. According to an example, the magnetic body M of the panel portion 220 may be magnetically attached to the edge portion 104 to maintain a coupled state.

Thereafter, as shown in FIG. 15, the mode is switched to use the mouse, which is referred to as a 'casual mode'.

The cover member 200 is separated and closely adhered to the lower portion of the mouse body 100.

When the cover member 200 is closely adhered, the magnetic body M of the panel portion 220 of the cover member 200 is magnetically attached to the edge portion 104 of the mouse body 100 so that a fixed state may be maintained.

While the mouse body 100 is gripped by hand, the power is turned on, and it is used after being connected to a computer wirelessly or by wired.

The cover member 200 moves along with the movement of the mouse body 100, the sensor 270 detects a position change according to the movement, transmits the position change to the mouse body 100, and also transmits the position change to the computer to function as an input device.

Although the embodiments of the present invention have been described in more detail with reference to the accompanying drawings, the present invention is not necessarily limited to these embodiments, and may be variously modified and implemented without departing from the technical spirit of the present invention.

What is claimed is:

1. A computer mouse comprising:

- a mouse body having a plurality of buttons on an upper portion and a scroll wheel formed between the plurality of buttons, and formed in an upwardly convex shape;
- a support unit including a plate formed on a lower portion and a cover formed on an upper portion, having a space formed inside, and formed in the middle of the space;
- a first driving unit moving the cover in a front and rear direction to perform an up-and-down operation;
- a second driving unit moving the plurality of buttons in a left and right direction to perform an open-and-close operation;

- a sub-button moving the cover backward to raise the cover and, at the same time, extending outward on both sides of the body or retracting into the body when the cover is moved forward; and

a third driving unit causing the sub-button to appear and disappear.

2. The computer mouse according to claim 1, wherein the support unit comprises:

- a plate member placed on an upper portion of the plate, with rack gears formed on both sides of an upper surface;

- a box portion protruding from an upper portion of the plate member, with a liquid crystal display unit and a plurality of setting switches formed on an upper surface of the box portion; and

- a scroll wheel formed in front of the box portion.

3. The computer mouse according to claim 1, wherein the first driving unit comprises:

- an operation axis having an upper end hinged to a bottom surface of the cover and a lower end hinged to a protrusion formed on an upper surface of the plate;

- a rotating portion including a boss protruding from the upper surface of the plate, a shaft coupled to the boss, and a main gear coupled to the shaft;

- a rack gear formed on an upper surface of the support unit in a longitudinal direction so as to engage with the main gear;

- a first pulley formed on a lower hinge axis of the operation axis;

- a second pulley formed on the shaft; and

- a belt connecting the first pulley to the second pulley.

4. The computer mouse according to claim 1, wherein the second driving unit comprises:

- a horizontal plate formed on a lower portion of the button and having a slide long hole; and

- a link having one end fitted to a first axis pin formed in front of the support unit and the other end fitted to a second axis pin inserted into the long slide hole of the horizontal plate, with a long hole formed to guide the movement of the first axis pin.

5. The computer mouse according to claim 1, wherein the third driving unit comprises:

- a connection bar formed on a side surface of the support unit and having a groove to accommodate a boss and a protrusion of the plate;

- a side link bar having one end hinged to the connection bar and having a slide groove of a certain length; and
- a side bar hinged to the other end of the side link bar, with the sub-button coupled to the side bar.

6. The computer mouse according to claim 5, further comprising:

- a folding unit rotating the sub-button to fold the sub-button in an up and down direction, wherein the folding unit includes

- a rod having one end inserted into the slide groove of the side link bar to be movable, and the other end hinged to the side bar;

- a slider coupled to a long hole formed in the side bar in the longitudinal direction and to which the other end of the rod is hinged; and

- an induction link having a long hole of which one end is fitted to an axis pin protruding from an upper surface of the slider, and the other end is hinged to the side bar.